Identity Shield: A Sequential Five-Gate Defense Against Biometric Identity Impersonation in an Aadhaar-Inspired Authentication Pipeline

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***Abstract*—National identity systems such as Aadhaar authenticate a resident by comparing a presented biometric to an enrolled record, but a legitimate ID number plus a spoofed or donated biometric defeats first-generation and score-fusion systems. This paper presents Identity Shield, an Aadhaar-inspired re- search prototype that secures authentication by placing five distinct hard-stop gates in series: credential verification via SHA-256 hashing, face presentation-attack detection (PAD) using a Random Forest classifier on color histograms, one-to-one verification using FaceNet embeddings, fingerprint PAD using a Random Forest classifier on gray-level statistics, and one-to-one fingerprint verification using ORB keypoint matching and brute-force Hamming distance. A request is au- thorized only if all gates are cleared, so a strong performance on one biometric cannot compensate for a weakness on another. The system is evaluated on a four-step web portal, end-to-end, using fully synthetic identities created from the SOCOFing fingerprint benchmark and a large-scale public face data set, against a script-based attack portfolio spanning credential, face, fingerprint, cross-modal, and system attacks. On a pre- liminary sample run, the unauthorized acceptance rate drops from 100:0the full pipeline, while attack categories are detected at 100:084:0layman's terms, analyzes how sequential conjuction reduces un- authorized acceptance, weighs the design choices affecting the fingerprint gates, and identifies the factors limiting ex- ternal interpretation of the results as production-grade accuracy.**

***Index Terms*—Aadhaar, biometric authentication, presenta- tion attack detection, multimodal biometrics, FaceNet, ORB, Random Forest, identity security.**

I. Introduction

ADHAAR is India's 12-digit unique identity number that is related to demographic data and to fingerprint, iris and face biometrics. It underlies banking know your customer checks, subsidy delivery, mobile SIM issuance, and e-governance services. Authentication is the matching of a new biometric sample to the stored record, and it takes place at a national level, as one centralized system contains the identity of over a billion residents [1].

Three characteristics of this scenario make security challenging. Firstly, biometrics are permanent: unlike a password, a fingerprint or a face cannot be reset once it has been compromised. Secondly, the data is centralized: a single breach or linkage attack can reveal many identities at once. Thirdly, biometrics are spoofable: artificial fingers, printed photographs, screen replays, and computer-generated faces can all be presented to a sensor and sometimes accepted [2].

A. Problem Statement

Consider an adversary that already possesses a valid 12-digit identity number, obtained either through a database leak or by reading it off a printed document, who presents forged or stolen biometric evidence. A system that checks only the number accepts the attempt outright. A system that combines the face and fingerprint scores into a fused score may still accept the attempt, since a compelling forgery on one modality could raise the combined score through the pass mark, despite the other modality being clearly wrong. The question this work asks is:

> *How effective is a sequential multi-layer authentication architecture in mitigating the risk of such an unauthorized access given a valid credential and the presence of incorrect or presentation-attacked biometric evidence?*

B. Motivation

The 2024 survey by Sadhya and Sahu [1] documents the threats to the Aadhaar framework, including fake biometrics, cloning, breaches, and linkage attacks, and discusses the preventive measures. The survey, descriptive in nature, is a proposal of no executable system and measurement of no defense. Our motivation is to implement the survey's threat list as a working, measurable pipeline that can be attacked and inspected from end to end by a student team using only public datasets and synthetic identities.

C. Contributions

The contributions of this work are:

1) A **sequential five-gate architecture** where credential verification, face PAD, face matching, fingerprint PAD and fingerprint matching are all hard-gates, with each attempt being audited, and the order determined by the backend and not the client

TABLE I
GLOSSARY OF TERMS USED IN THIS PAPER.

Term	Meaning in this paper
Spoof / presentation attack	Showing the sensor something fake, such as a printed photo, a screen replay, a generated face, or an altered fingerprint
PAD	Presentation-attack detection: deciding whether a sample is genuine or fake before asking whose it is
Verification (1:1)	Checking a sample against the single record claimed by the identity number
Hash	A fixed-length fingerprint of data that cannot be turned back into the original
Embedding	A list of numbers that describes a face so that similar faces get similar lists
Histogram	A count of how many pixels fall in each brightness range
Random Forest	Many small decision trees that vote on a yes/no question
Hard stop	A check whose failure ends the whole attempt immediately
UAR	Unauthorized acceptance rate: the share of attacks that were wrongly accepted

2) A **decision model in natural language** explaining why demanding passage through all gates is necessary to prevent spoofing and why aggregation of scores is not informative (Section III).
3) A **synthetic identity dataset** created using SOCOFing [3] and a public face dataset [4], so that no real Aadhaar record is used.
4) A **working web prototype** with a four-step verification portal and a research mode for attack analysis (Section IV).
5) A **scripted attack suite** of 25 attack scenarios plus one genuine baseline in five families, and an **ablation over four pipeline configurations** (Section V).
6) A **preliminary evaluation** together with an explicit list of the limits of the evidence (Section VI).

D. Organization

Section II explains the background techniques and reviews related work. Section III describes the architecture and the five gates. Section IV walks through the implementation and the user interface. Section V defines the experiments, Section VI reports and discusses results, Section VII covers security, privacy, and ethics, Section VIII lists limitations and future work, and Section IX concludes.

II. BACKGROUND AND RELATED WORK

This section explains, without formulas, each technique that the prototype relies on, and then reviews related work and its limitations. Table I gives a short glossary.

A. Aadhaar Authentication and Its Attack Points

In the Aadhaar flow, a user provides the identity number along with a fingerprint or a face. A capture device reads and encrypts the biometric and an authentication service sends the query to the central authority, the central database compares the sample to the enrolled template and a yes/no answer comes back. Sadhya and Sahu [1] review the data-security and privacy risks of this arrangement and identify the main threats as fake biometrics, cloning, breaches and linkage attacks.

TABLE II
ATTACK ENTRY POINTS AND COVERAGE BY THE PROTOTYPE.

Entry point	Gates	Coverage
Stolen ID number	1, 3, 5	The number alone does not pass the biometric gates
Fake fingerprint	4, 5	Altered-print PAD and matching; see limits
Fake face	2, 3	Synthetic-image PAD and matching
Database breach, replay	none	Credentials stored as hashes only; replay is not addressed

From the point of view of an attacker, there are four entry points: a stolen identity number and demographics from a breach or phishing; a fake fingerprint made from lifted or stolen fingerprint images; a fake face such as a printed photo, a screen replay, or a deepfake; and the central database or a replay of captured biometric data. Identity Shield addresses the first three at authentication time and stores credentials only as hashes. Table II states what the prototype does and does not cover.

B. Protecting Credentials with a Hash

A cryptographic hash such as SHA-256 turns any input into a 64-character fingerprint with three useful properties. The same input always gives the same output. Changing even one digit of the input produces a completely different output. And, in practice, the output cannot be run backwards to recover the input. Storing only the hash of an identity number means that a stolen database copy does not directly reveal the numbers. There is a caveat that we return to in Section VII: a 12-digit number has only a trillion possible values, which is small enough that an attacker holding unsalted hashes could try them all.

C. Face Verification with Embeddings

Deep-learning approaches beat hand-designed features for deciding whether two photographs show the same person, using a pipeline of detecting the face, aligning it, converting it to a learned representation, and comparing representations. FaceNet [5] refined this by training a network to turn each face into a list of 128 numbers, called an embedding. During training the network sees triples of photographs: an anchor, another photograph of the same person, and a photograph of a different person. It is rewarded for placing the two photographs of the same person closer together than the photograph of the stranger, by a safety margin. After training, faces of one person cluster together and different people spread apart.

To verify, the system converts the presented face and the enrolled face to embeddings and measures how far apart they are. A common measure is cosine distance, which looks at the angle between the two lists: a distance near zero means

almost identical, and larger values mean less similar. If the distance is below a threshold chosen for the model, the faces are declared a match. We use FaceNet through the DeepFace library. The limitation of an embedding matcher on its own is that it answers "is this the same person?" and not "is this a real presentation?", so a good photograph of the enrolled person can pass.

D. Face Presentation-Attack Detection with Color Histograms

PAD methods look for physical or statistical traces that separate skin from printed or displayed media. Boulkenafet *et al.* [6] show that color and texture statistics can separate genuine from spoofed faces. Faces generated by style-based adversarial networks add another class of fake that looks natural to a human viewer.

A color histogram is one of the simplest such descriptors. For each color channel (red, green, blue) the pixel values from 0 to 255 are grouped into 32 equal ranges, and the share of pixels falling in each range is recorded.The three channels combined provide 96 numbers that describe the overall color distribution of the face image regardless of its size. A printed photograph, a screen and a generated image typically have slightly different color distributions than a camera photograph of a real face, and a classifier can learn this difference.

E. Random Forests

A Random Forest [7] is a set of decision trees. Each tree is a small flowchart that asks binary questions of the input values (like "is the percentage of bright red pixels over some threshold?"), and each tree was trained on a slightly different random subset of the training data. When a test example is given, each tree votes on whether the example is genuine, and the proportion of votes for genuine is the probability. Forests are fast to train and run, have good performance with small feature vectors, do not require graphics cards, and are easy to interpret. That is why we use them for the two PAD gates. Our forests have 100 trees with at most 10 levels each.

F. Fingerprint PAD and Matching

Artificial fingers made of gelatin or silicone were shown to be accepted by commercial scanners [2], Convolutional networks give strong liveness results [8] at higher computational cost than simple statistical descriptors.

For matching, the standard approach is to extract minutiae, the points where ridges end or split. Our prototype uses ORB [9] as a lighter baseline. ORB works in three steps. First, it finds corner-like points that stand out in the image. Second, for each point it measures the direction in which brightness is concentrated, so that the surrounding patch can be rotated to a standard orientation, which makes the description robust to rotation of the finger.Third, it represents the patch as a string of 256 bits, each bit recording the result of a brightness comparison between two selected pixel locations. Two such strings are compared by counting the number of differing bits which is called the Hamming distance. Few differing bits mean similar patches.A brute-force matcher compares every descriptor of the probe print with every descriptor of the enrolled print, and the number of pairs that are close enough is the match count. ORB keypoints are not minutiae, so this stage is a baseline and not a replacement for an AFIS-grade matcher.

TABLE III
FUSION LEVELS IN MULTIBIOMETRIC SYSTEMS.

Level	What is combined	Security note
Sensor	Raw signals	Needs synchronized, co-registered sensors
Feature	Joined feature lists	Face and fingerprint features live in incompatible spaces
Score	Normalized match scores	Strong evidence in one modality can compensate for another
Decision / sequential	Per-modality accept or reject	No compensation; each modality must succeed

G. Multimodal Fusion and the Compensation Problem

Multibiometric systems are classified by the level at which evidence is combined (Table III) [10]. The most common approach, score-level fusion, converts each matcher's output to a common scale and takes a weighted average.

The security weakness of score fusion is *compensation*. As an illustration, not a measurement, suppose the credential counts for 20% of the final score, the face for 50%, and the fingerprint for 30%, and the pass mark is 0.70. An attacker with a valid credential (score 1.0) and an excellent forged face (score 0.96) contributes 0.20 plus 0.48, which is already 0.68. Even a clearly wrong fingerprint scoring only 0.20 adds 0.06 and brings the total to 0.74, above the pass mark. Access is granted although the fingerprint is wrong. Fig. 1 shows the same effect geometrically: the combined-score rule accepts the whole shaded region to the right of its decision line, including points with a very low fingerprint score, whereas a rule that demands that each modality pass on its own accepts only the square in which both scores are high.

H. Limitations of Existing Approaches and the Need for the Proposed Model

Table IV summarizes related work and the limitation of each piece that matters here. Each component technique is individually established. Face matchers [5] have no notion of presentation attacks. Face and fingerprint PAD methods [6], [8] are evaluated one modality at a time, separately from identity binding. Score fusion [10] combines modalities but admits compensation. The gap is therefore not a missing algorithm but a missing *composition*: an end-to-end pipeline in which credential, liveness, and identity checks for both modalities must all hold, and which can be attacked and measured as a whole. Table V contrasts the reference survey with this work.

III. System Design

A. Design Principles

Four principles guided the design. *Every gate is a hard stop*: a failure ends the attempt immediately, so no later success can

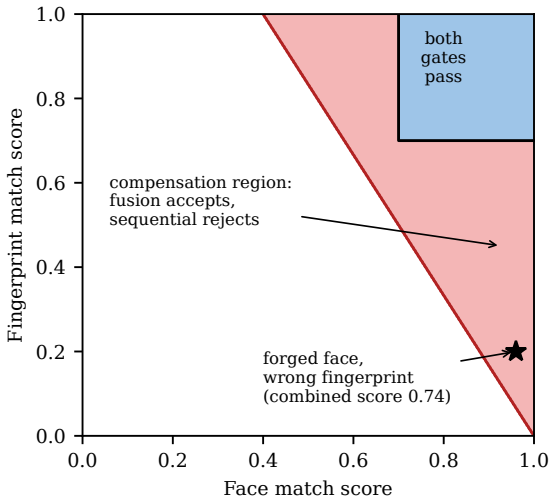


Fig. 1. Acceptance regions in the face–fingerprint score plane for a valid credential. The weights, the pass mark of 0.70, and the per-modality thresholds of 0.70 are illustrative. This is an analytic illustration, not measured data.

TABLE IV
RELATED WORK AND LIMITATION RELEVANT TO THIS STUDY.

Work	Focus	Limitation here
Sadhya and Sahu [1]	Aadhaar threat survey	Descriptive; no executable system or measurements
Schroff *et al.* [5]	Face verification	Match identity only; no PAD
Boulkenafet *et al.* [6]	Face PAD (color texture)	Face only; no link to identity or fingerprint
Matsumoto *et al.* [2]	Gummy-finger attacks	Demonstrates the threat; not a defense pipeline
Nogueira *et al.* [8]	Fingerprint PAD	Single modality; CNNs cost more compute
Ross and Jain [10]	Score-level fusion	Compensation between modalities

compensate. *Cheap checks come first*: hashing and histogram classification run before the deep face network, so malicious traffic is dropped before the expensive step. *The server enforces the order*: the client cannot skip a step, and a request with a missing input is blocked. *All logged*: each attempt, accepted or blocked is recorded with the result of every gate that ran.

B. System Architecture

Fig. 2 shows the three tiers of the prototype. The web client collects the identity number, the face image and the fingerprint image in four steps and displays the result. The backend is a FastAPI app served by Uvicorn. It receives the request and passes it to an orchestrator that runs the five gates in a fixed order and stops at the first failure. Each gate accesses its own resource in the data tier (the SQLite database for credential hashes and demographics, the trained Random Forest models for the two PAD gates, the FaceNet model and the enrolled face for face matching, and the enrolled fingerprint for ORB matching). Finally the response builder returns the decision, names the failing gate if any, produces an identity card on success, and writes an audit record.

TABLE V
REFERENCE SURVEY VERSUS THIS WORK.

Sadhya and Sahu [1]	This work
Security and privacy survey	Executable prototype
Describes threats and measures	Controlled attack simulation
Broad Aadhaar ecosystem	Face and fingerprint, five sequential gates
Descriptive	Quantitative, gate-wise evaluation

TABLE VI
GATE SUMMARY.

Gate	Method	Feature / input	Decision
1	SHA-256 lookup	ID to 64-character digest	valid / invalid
2	Random Forest	96-number color histogram	live / spoof
3	FaceNet (DeepFace)	128-number embedding	match / non-match
4	Random Forest	34-number grayscale statistics	genuine / spoof
5	ORB + BFMatcher	keypoint descriptors, 15 good matches	match / non-match

C. The Five-Gate Pipeline in Detail

Fig. 3 expands the orchestrator into its five gates, listing for each the input it consumes, the features it computes, the method it applies, the condition for passing, and how a failure is reported. The gates are ordered so that credential and image-statistics checks, which are cheap, come before the deep face network, and so that each presentation-attack check comes before the matching check for the same modality: a sample must first look real before the system asks whose it is.

Table VI lists the same information compactly, together with the kind of model used.

D. Threat Model

The attacker knows or can guess a valid identity number and controls the biometric evidence submitted to the verifier, but cannot modify server code or the database. In scope are: invalid or altered identity numbers; another person's face or fingerprint; synthetic or fake face images; altered fingerprint images; and cross-modal combinations in which one modality is genuine and the other is not. Out of scope are physical sensor attacks beyond static images, network interception, and compromise of the host. These are discussed under limitations in Section VIII.

E. Synthetic Identity Dataset

No real Aadhaar record is used. Faces come from the Biggest Gender/Face Recognition Dataset [4] and fingerprints from SOCOFing [3], which has 6,000 real prints from 600 subjects together with more than 55,000 synthetically altered copies. Fig. 4 shows how identities are built.

In words: faces are labeled by a pretrained FairFace-based gender classifier [11], which returns a label and a confidence, while fingerprints are labeled from the male or female marker

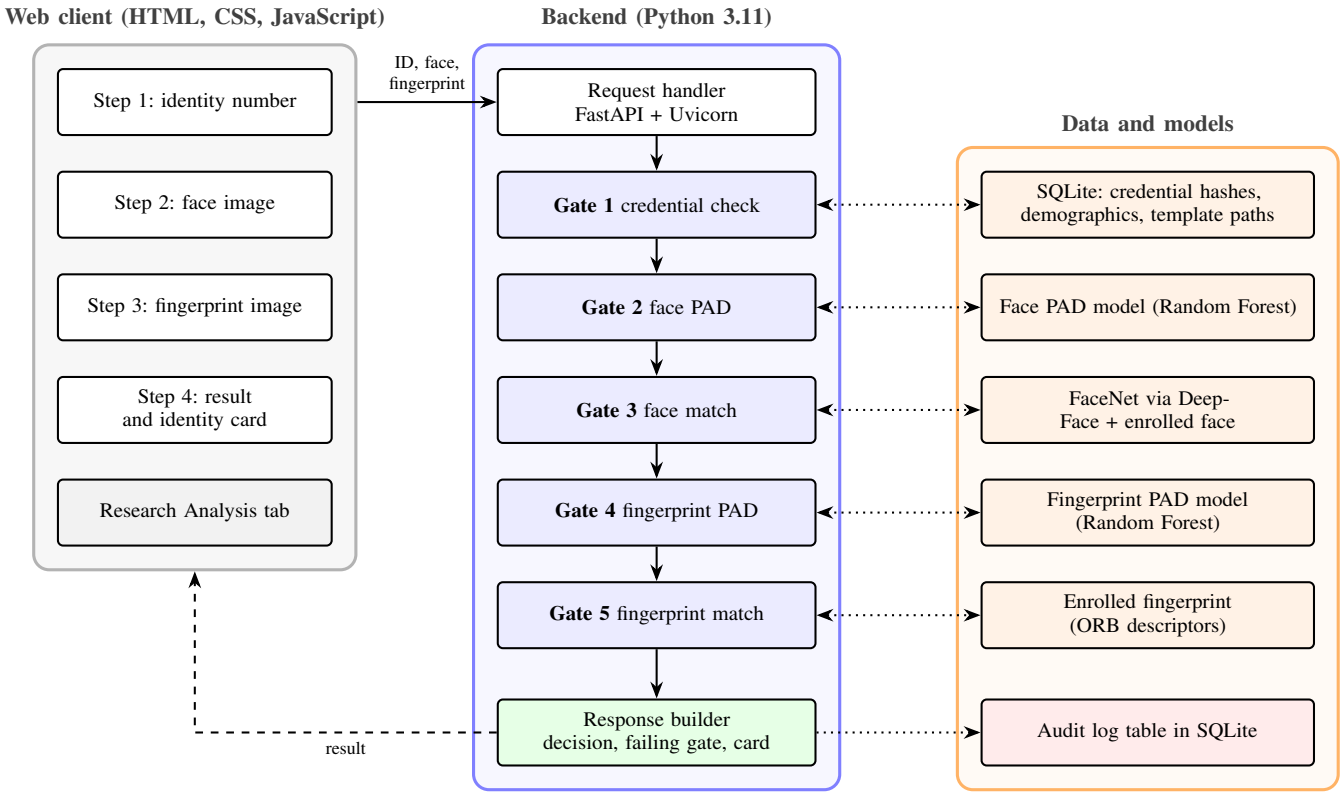


Fig. 2. System architecture of Identity Shield. The client collects inputs, the backend orchestrator runs the five gates in a fixed order, and each gate reads its own model or record in the data tier. Dotted lines are reads or writes; the dashed line returns the result to the client.

in the SOCOFing filename. One face and one fingerprint of the same gender are combined to form an identity, and the face files are renamed with a male or female prefix. Each identity then receives a random Indian name, age between 18 and 80, gender, and address. Finally, a unique 12-digit number is issued, subject to the condition that no digit appears more than twice, and only the SHA-256 hash of the number is stored. This produces a table loaded into SQLite with the hash, the template paths, and the demographics. The credentials, templates, and demographics are all in separate tables, limiting the information available in the event of a leak.

F. Gate 1: Credential Verification

The submitted string is first checked to be exactly 12 digits. It is then hashed with SHA-256 and the hash is looked up in the credential table with a parameterized query. A parameterized query sends the user's text to the database as plain data and never as part of the command, so injected database commands are treated as an ordinary string that is simply not found. The gate passes if the hash is present; plaintext identity numbers are never stored. Because hashing is nearly instantaneous, this gate removes random guesses, one-digit changes to a valid number, and malformed input at almost no cost.

G. Gate 2: Face PAD

The face image is resized to 128 by 128 pixels and converted to the 96-number histogram described in Section II. The Random Forest returns the probability that the image is a real face, and the gate passes if that probability is high enough, with the forest's default decision point of 0.5. The classifier is trained to separate genuine face images from fake or synthetic ones. This gate is a *static-image* detector: it examines image statistics and not physical liveness, so it cannot tell a live face from a replay of a plausible image.

H. Gate 3: Face Verification

Both the presented face and the enrolled face are converted to FaceNet embeddings through the DeepFace library, and the distance between them is compared with the library's threshold for the FaceNet model. A distance below the threshold means the two faces are judged to be the same person. The library's default for FaceNet with cosine distance is 0.40. This gate catches another person's face, including a live impostor and a generated face that happens to pass Gate 2.

I. Gate 4: Fingerprint PAD

The fingerprint is converted to grayscale and resized to 96 by 96 pixels. The feature is a list of 34 numbers: a 32-range brightness histogram plus the average brightness and its spread. A second Random Forest of 100 trees and depth 10 is trained with the real SOCOFing prints as the genuine class and the altered SOCOFing copies as the spoof class. The gate passes when the genuine probability is high enough. Altered prints are a proxy for spoofs: they reproduce software distortions

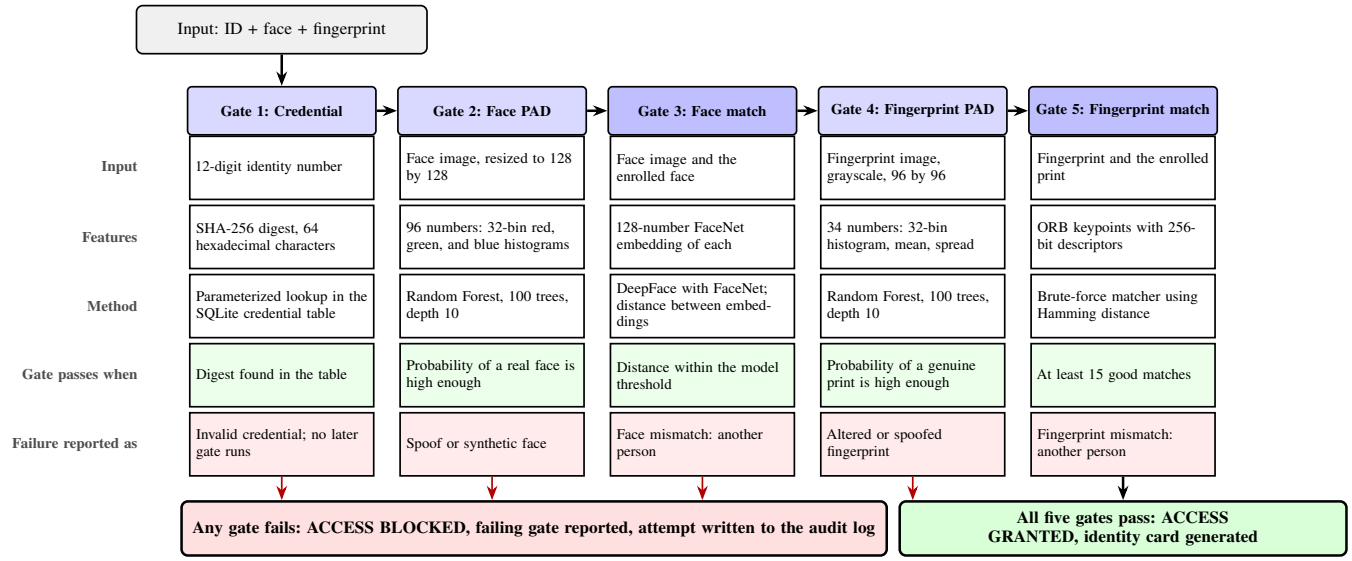


Fig. 3. The five-gate pipeline. Each column is one gate; every gate is a hard stop. The green row states the passing condition and the red row states how a failure is reported. Gates 2 and 4 look at whether a sample is real; Gates 3 and 5 look at whose it is.

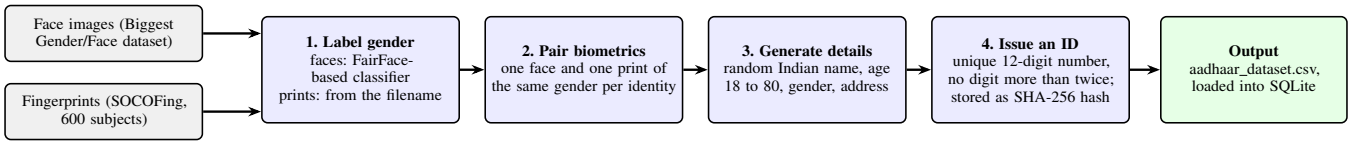


Fig. 4. Construction of the synthetic identity dataset. The face–fingerprint pairing is a construction of this study and not ground truth about one person.

such as obliteration, central rotation, and Z-cuts, but not the optics of a real gelatin or silicone finger.

J. Gate 5: Fingerprint Matching

ORB keypoints and binary descriptors are extracted from the presented and the enrolled print and compared with the brute-force matcher under Hamming distance. A pair of descriptors counts as a good match if their bit strings are similar enough. The gate passes only if there are at least 15 good matches. Fifteen is a fixed count, and it is the reason this gate can reject another person's fingerprint: an unrelated print will usually produce few close pairs.

K. Why Sequential Conjunction Limits Unauthorized Acceptance?

The decision rule of the whole system can be stated in one sentence: the request is accepted if and only if the credential, the face PAD, the face match, the fingerprint PAD, and the fingerprint match all pass, and a later gate runs only if every earlier gate has passed. Three consequences follow, and they are the reasoning behind the architecture.

First, the chance that an attack gets through can never exceed the chance that it passes any single gate. If an attack fools the fingerprint PAD only one time in ten, then at most one attack in ten can be accepted, however well it fools the other gates. This is the converse of score fusion, where a good score on one modality may push the totals over the pass mark (Fig. 1).Second, if the gates behave independently for a given attack, the probability of getting through all five is the product of the probabilities of passing each, so adding gates multiplies the obstacles. Independence is an idealization here, because the two face gates share an input image and the two fingerprint gates share another, so their outcomes tend to be related; the product should be read as an optimistic estimate and not a guarantee. Third, because gates run in order and stop at the first failure, the average cost of an attempt is dominated by how many requests survive to the expensive gates. Putting cheap, discriminating gates first means that most malicious requests never reach the deep face network. We did not measure latency in this study, so this is a design argument and not a result.

L. Usability Cost of Hard Stops

The same logic applies to genuine users who need to pass each gate. Hence, small error rates may sum up to a considerable one. As an example, let's imagine that each of the 5 gates incorrectly rejects 2 out of 100 genuine users. Then, approximately, 10 out of 100 would be rejected on average in the whole system. Therefore, false rejections must also be taken into account and the threshold for each of the gates should be tuned in accordance with the trade-off between unauthorized acceptances and false rejections.

M. Audit Logging

Every attempt is stored with the result of each gate that ran, the failing gate if any, and the attack label when one is

TABLE VII
GATES EXECUTED IN REPRESENTATIVE SITUATIONS (FROM THE DESIGN).

Situation	Gates run	Outcome
Genuine user, all inputs correct	1, 2, 3, 4, 5	Access granted
Invalid or altered ID	1	Blocked at Gate 1
Valid ID, another person's real face	1, 2, 3	Blocked at Gate 3
Valid ID, generated face	1, 2	Blocked at Gate 2
Right face, another person's print	1 to 5	Blocked at Gate 5
Right face, altered print	1 to 4	Blocked at Gate 4
Missing input	none	Blocked by order enforcement

TABLE VIII
REPOSITORY ORGANIZATION.

Component	Role
`authentication/credential`	Gate 1: hash and lookup
`authentication/face_pad`	Gate 2: face PAD
`authentication/face`	Gate 3: FaceNet verification
`authentication/fingerprint_pad`	Gate 4: fingerprint PAD
`authentication/fingerprint`	Gate 5: ORB matching
`authentication/fusion`	Orchestrator, hard-stop conjunction
`api/`, `frontend/`, `database/`	FastAPI, web client, SQLite
`models/`	Trained PAD models

supplied in research mode. This provides an audit trail, which the survey notes is missing from many deployments [1], and it is the source for the per-gate analysis in the research tab.

N. Request Lifecycle

Fig. 5 follows one verification request through the system. The client submits the three inputs, the orchestrator runs Gate 1 against the database, and if it passes the orchestrator runs Gates 2 to 5 in turn, each reading its own model or enrolled template. After every gate the result returns to the orchestrator, which stops at the first failure. Whatever the outcome, an audit record with all gate results is written, and the client receives the decision, the name of the failing gate if there was one, and the identity card on success.

O. Worked Examples

Table VII walks through seven situations and states which gates run and where the attempt ends. These follow from the design rules above, and they show two properties of the hard-stop order: attacks are stopped at the earliest gate that can see them, and later gates are never reached, so their cost is never paid.

P. Design Decisions and Trade-offs

Several choices were made for simplicity and explainability, and each has a cost.

Random Forests for PAD. Small forests on short feature lists train in seconds on a laptop, need no graphics hardware, and can be explained to a non-specialist. The cost is accuracy: global color and brightness statistics cannot capture fine texture such as moiré patterns, print dots, or ridge flow, which is where state-of-the-art PAD methods gain their strength.

FaceNet through DeepFace. A pretrained embedding model gives strong face matching without training our own network, and the library handles detection, alignment, and thresholds. The cost is a dependence on a large model that is loaded in memory and is the slowest step of the pipeline.

ORB for fingerprints. ORB is free of patents, quick to compute, and available in OpenCV. The cost is that it was designed for natural images, not for ridge patterns, so it is a baseline.

Hard stops over weighted scores. Hard stops are easy to reason about and prevent compensation, but they multiply the chances of falsely rejecting a genuine user, as discussed above. A production system might use hard stops for the presentation-attack gates and a calibrated rule for matching.

SQLite and FastAPI. A single-file database and a lightweight web framework make the prototype easy to run and inspect on any machine, which suits an academic project but not a national-scale service.

IV. IMPLEMENTATION

A. Software Stack and Organization

The backend runs on Python 3.11 with FastAPI and Uvicorn. FaceNet is run through TensorFlow and DeepFace, the Random Forests are built with scikit-learn, and fingerprint processing uses OpenCV. The database is SQLite, and the client is a framework-free page written in HTML, CSS, and JavaScript. A batch script starts the server on the local loopback address, loads the PAD models and the DeepFace model into memory, and opens the web client. Table VIII shows how the repository is organized: one module per gate and one orchestrator. Despite the name of the orchestrator module in the repository, it performs a hard-stop conjunction and not score fusion.

B. Verification Portal Walkthrough

The portal has two tabs, Verification Portal and Research Analysis, and a footer stating that the system is an academic research prototype, is not affiliated with UIDAI or any government authority, and shows only synthetic data. The verification flow has four steps shown in a progress bar, and Figs. 6 and 7 show them.

Step 1, Identity. The user enters the 12-digit identity number. A line under the field tells the user that the number is securely hashed before verification, and the Continue button moves to the next step. Gate 1 is evaluated using this value.

Step 2, Face. The user positions a face in the frame or, for testing, uploads a face image. The page explains that face verification checks whether the presented face corresponds to the registered identity. Behind this single step, Gates 2 and 3 are applied to the uploaded image, so the portal shows the user one face step while the backend performs a liveness check and a match.

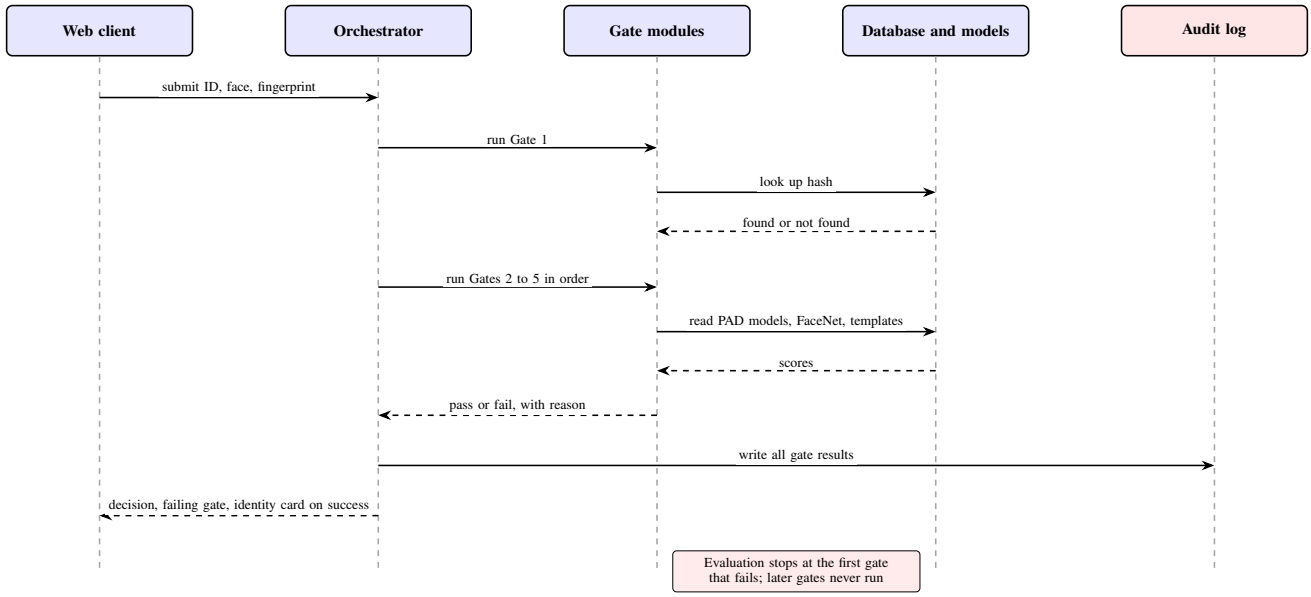


Fig. 5. Lifecycle of one verification request. Solid arrows are requests, dashed arrows are replies.

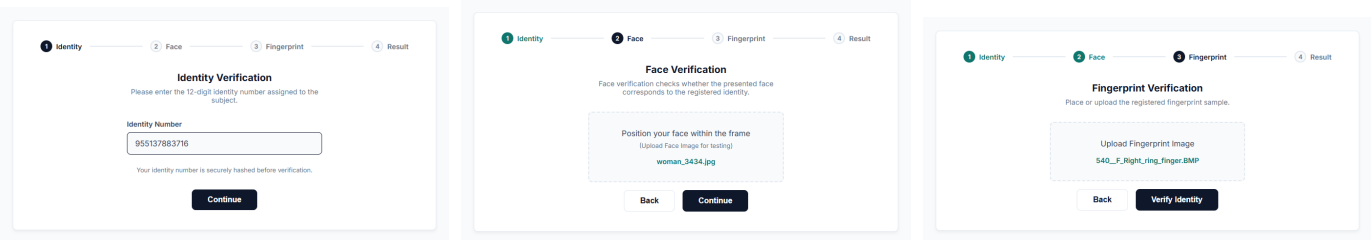


Fig. 6. Verification portal, steps 1 to 3 (left to right): identity number entry with a note that the number is hashed before verification; face upload (a test image, `woman_3434.jpg`); fingerprint upload (`540__F_Right_ring_finger.BMP`) with the Verify Identity button.

Step 3, Fingerprint. The user places or uploads the registered fingerprint sample, and the Verify Identity button submits all three inputs. Gates 4 and 5 are applied to the fingerprint image.

Step 4, Result. The result page (Fig. 7) lists four checks, Identity Credential, Face Verification, Fingerprint Verification, and Biometric Consistency, each with a Verified mark. Below them a green banner states that all required identity and biometric checks have been completed successfully, and a card shows the matched synthetic record: a photograph, name, age, gender, address, and the identity number in groups of four digits. The card is labeled demo data and not an official record, and a collapsible Verification Details section is available. The portal groups the five gates into these user-facing checks; Biometric Consistency is a closing row that summarizes the agreement of the face and fingerprint results.

C. Research Analysis Mode

The second tab is used for attack experiments. In this mode the tester uploads deliberately wrong or spoofed inputs, and the system maps each failure to an attack category and to the gate that blocked it. The same per-gate results are written to the audit log, so the charts in Section VI can be regenerated from the log of a test run.

D. Reproducing the System

The prototype is designed to run on a single computer. A setup step installs the Python dependencies, a dataset-preparation step builds the synthetic identity table from the two public datasets, a training step builds the two PAD models, and the batch script then starts the server on the local address and opens the portal. A tester can enter the printed identity number of a synthetic user, upload that user's enrolled face and fingerprint, and watch all four steps pass; changing any one input reproduces the corresponding attack in Table X. Because the datasets are public and the identities are generated, no restricted data is needed to repeat the work.

V. Experimental Setup

A. Pipeline Configurations

To measure the contribution of each layer, the same attacks are run against four configurations (Table IX). Configuration C contains both matchers but no PAD gates, so it represents a conventional design that checks the identity number and then matches the biometrics. The difference between C and D therefore isolates the value of presentation-attack detection.

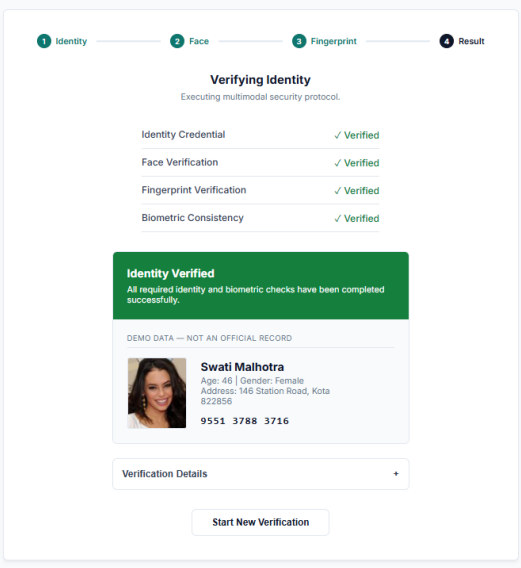


Fig. 7. Verification portal, step 4: all checks verified and the digital identity card generated from the matched synthetic record.

TABLE IX
PIPELINE CONFIGURATIONS EVALUATED.

Config.	Layers used	Gates active
A	Credential only	1
B	Credential and face verification	1, 3
C	Credential, face, and fingerprint verification	1, 3, 5
D	Full Identity Shield	1 to 5

B. What Each Configuration Isolates

The four configurations form a ladder, and each step answers one question. Moving from A to B asks how much a face matcher adds on top of the identity number. Moving from B to C asks how much a second modality adds. Moving from C to D asks how much presentation-attack detection adds when both matchers are already present. Because the attack suite is identical across configurations, any change in the unauthorized acceptance rate between two neighbouring rows can be attributed to the layers that were added, and not to a change in the attacks. This is the usual logic of an ablation study, and it is the reason the experiment compares configurations instead of reporting a single number for the full system.

C. Attack Suite

The suite contains 25 attack scenarios and one genuine baseline, and each scenario is run for two enrolled users. Fake-face samples are drawn from a folder of generated faces. Table X lists the families and the gate expected to stop each one by design. The expected blocking gate is a design expectation, not an observed result.

TABLE X
ATTACK FAMILIES AND EXPECTED BLOCKING GATE.

Family	Scenario	Blocked at
Credential	Invalid ID, or one digit of a valid ID changed	Gate 1
Face	Valid ID, another person's face	Gate 3
Face	Synthetic or fake face image	Gate 2
Fingerprint	Valid ID, another person's print	Gate 5
Fingerprint	Altered or latent print	Gate 4
Cross-modal	Correct face, wrong fingerprint	Gate 5
Cross-modal	Wrong face, correct fingerprint	Gate 3
Cross-modal	Wrong face, wrong fingerprint	Gate 3
Robustness	Heavily blurred target face	Gate 2 or 3
System	Step bypass, missing input	Order enforcement
Baseline	Genuine user, all inputs correct	Accepted

D. Metrics

The primary metric is the unauthorized acceptance rate (UAR): the percentage of attack attempts that the configuration wrongly accepted. It is computed over attack attempts that carry a valid credential, which is the threat posed by the research question. For the full system we also report the attack detection rate (ADR) per attack family, which is simply one hundred percent minus the unauthorized acceptance rate within that family.

E. Planned Reporting Protocol

Standard PAD reporting under ISO/IEC 30107-3, and the usual match-level error rates, are not computed in this study and are defined here so that they can be reported in the completed evaluation. The attack presentation classification error rate (APCER) is the share of presentation attacks wrongly treated as genuine. The bona fide presentation classification error rate (BPCER) is the share of genuine presentations wrongly treated as attacks. Their average is the average classification error rate. The false acceptance rate and false rejection rate are the corresponding shares for the matcher gates: impostors wrongly matched, and genuine users wrongly rejected. Each rate should be given together with a confidence interval, for example a Wilson interval, because with a few dozen attack runs per configuration the sampling uncertainty is large.

VI. RESULTS AND DISCUSSION

A. Effect of Each Layer

Fig. 8 and Table XI show the unauthorized acceptance rate from a sampled run of the attack set. Credential-only authentication accepts every attack that carries a valid ID (100.0%). Adding face verification lowers the rate to 76.5%, adding fingerprint verification lowers it to 51.5%, and enabling the two PAD gates lowers it to 9.0%. Relative to configuration C, the conventional design, the full pipeline removes about 82.5% of the accepted attacks (a drop from 51.5 to 9.0); relative to configuration B the figure is about 88.2%.

The largest single step is from C to D, which is consistent with the design: fake faces and altered prints are genuine-looking inputs that matchers alone are not built to reject, whereas the PAD gates examine the sample itself. This

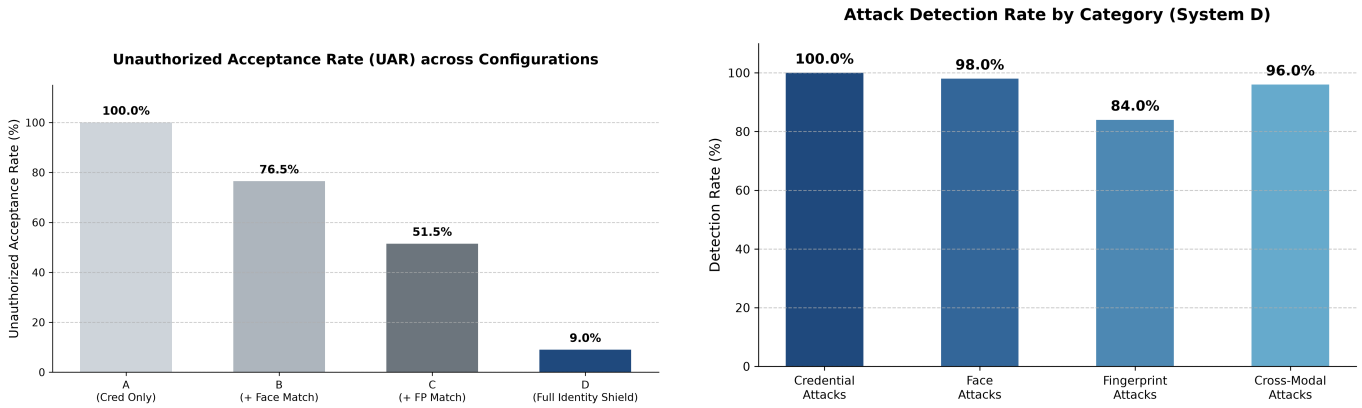


Fig. 8. Results of the sampled run. Left: unauthorized acceptance rate across the four configurations (lower is better). Right: attack detection rate by category for the full system, D (higher is better). The run should be repeated on the full attack set before final claims.

TABLE XI
UNAUTHORIZED ACCEPTANCE RATE BY CONFIGURATION (SAMPLED RUN).

Config.	UAR (%)	Reduction vs. A (points)
A: credential only	100.0	none
B: plus face match	76.5	23.5
C: plus fingerprint match	51.5	48.5
D: full five-gate	9.0	91.0

TABLE XII
ATTACK DETECTION RATE BY CATEGORY, FULL SYSTEM D (SAMPLED RUN).

Attack category	Detection rate (%)
Credential attacks	100.0
Face attacks	98.0
Cross-modal attacks	96.0
Fingerprint attacks	84.0

is consistent with the hypothesis that combining credential verification, matching, and PAD across both modalities reduces unauthorized acceptance compared with credential-only or single-biometric authentication in this controlled experiment.

B. Detection by Attack Category

The right panel of Fig. 8 and Table XII show that the full system detects all credential attacks (100%), 98% of face attacks, and 96% of cross-modal attacks, but only 84% of fingerprint attacks. Credential attacks are deterministic: an unknown hash is simply not found. Face attacks are stopped by Gate 2 when the image is synthetic and by Gate 3 when it shows another person. Cross-modal attacks are stopped because both modalities are checked against the same enrolled identity: a correct face with a wrong print reaches Gate 5, and a wrong face is rejected earlier at Gate 3.

C. Comparison with the Survey Baseline

Table XIII compares, qualitatively and from the system design, a baseline that checks the identity number and matches biometrics (the situation described by the survey) with Identity Shield. Both block a wrong face and a wrong fingerprint, because a matcher rejects another person's sample. Only Identity Shield blocks a fake face or a fake fingerprint that belongs to the right person or looks plausible, because only it has PAD gates.

TABLE XIII
QUALITATIVE COMPARISON FROM SYSTEM DESIGN.

Attack	ID + matching	Identity Shield
Wrong face	blocked	blocked
Wrong fingerprint	blocked	blocked
Fake face (photo or GAN)	not blocked	blocked
Fake fingerprint	not blocked	blocked

D. Why Fingerprints Are the Weakest Category

The fingerprint gates are the weakest by design choice. Gate 4 uses 34 global grayscale statistics, which cannot see ridge-level structure, and its spoof class is made of software-altered images and not of physical casts. Gate 5 uses ORB keypoints, which respond to image corners and texture and not to minutiae, so impostor prints with similar texture can accumulate enough matches. The 16% of fingerprint attacks that passed is consistent with the limits of these two baselines, although we have not isolated which gate let each one through. Replacing them with a CNN-based PAD [8] and a minutiae matcher is the clearest route to improvement.

E. Residual Risk

A rate of 9.0% for the full system is a clear improvement over the baselines but is far from an acceptable absolute figure for a national identity service. We therefore present it as evidence that the ordering and hard-stop design work, not as evidence that the current gates are strong enough to deploy.

F. Failure Modes and Mitigations

Table XIV lists, gate by gate, what can still fool the current implementation and what would reduce the risk. It complements

TABLE XIV
FAILURE MODES OF EACH GATE AND POSSIBLE MITIGATIONS.

Gate	What can fool it	Mitigation
1	A leaked valid number; offline guessing of unsalted hashes	Keyed hashing; rate limiting; never rely on the number alone
2	Replay of a plausible image; high-quality prints and screens	Video or challenge–response liveness; texture-aware CNN PAD
3	A look-alike or a very good photograph of the person	Pair with PAD, as done here; stricter threshold
4	Physical casts with realistic texture	Sensor-level liveness; CNN PAD trained on physical spoofs
5	Impostor prints with similar texture	Minutiae-based matching; larger match count

the measured results by showing where the next improvements should go.

G. Threats to Validity

- **Sampled run.** The reported numbers come from a sampled run and are not yet confirmed on the complete attack set; no confidence intervals are given.
- **Synthetic pairing.** Face and fingerprint images come from different datasets and are paired arbitrarily, so the identities are not ground truth about one person.
- **Static images.** Gate 2 sees one frame and cannot distinguish a live face from a high-quality replay; video or challenge–response liveness is needed.
- **Proxy spoofs.** Altered SOCOFing prints and generated face images are not physical presentation attack instruments.
- **Missing metrics.** False acceptance and false rejection rates, processing time, and APCER and BPCER are not measured, so the usability cost of the extra gates is unknown.

VII. SECURITY, PRIVACY, AND ETHICS

Credential storage. Storing only a hash of the identity number avoids holding plaintext numbers. However, the identity space has only about a trillion values, so an unsalted SHA-256 hash could be recovered by trying every possible number on commodity hardware if the database leaked. A deployment should use a keyed hash, in which a secret held outside the database is mixed in, or a deliberately slow password-hashing function.

Data minimization. Credentials, templates, and demographics are held in separate tables, which reduces linkage risk compared with a single record that joins them [1]. India's Digital Personal Data Protection Act, 2023 places purpose-limitation and safeguard obligations on those who process personal data. The prototype uses no personal data, but a production version would require a legal review against these obligations.

Fairness and dataset bias. Face recognition and face PAD models can perform unevenly across gender, age, and skin tone, which in an identity system would mean unequal rates of wrongful rejection. FairFace [11] was built because earlier face datasets were unbalanced. Our gender labels for the face images come from a classifier and are therefore noisy, the face dataset is not demographically balanced by construction, and we did not measure error rates per group. Any deployment-oriented follow-up must report error rates per demographic group and not only the overall average.

Replay and transport. The prototype does not defend against replay of a captured request or against interception on the network. A production system would add session nonces, transport encryption, and device attestation.

Ethics and scope. All identities are synthetic and the portal says so on every page. The prototype is not affiliated with UIDAI, uses no UIDAI interface, and its attack simulation is a controlled research exercise on fabricated data.

VIII. LIMITATIONS AND FUTURE WORK

The main limitations are the sampled evaluation, the static-image PAD, the baseline ORB matcher, the arbitrary pairing of face and fingerprint images, and the absence of replay protection. Future work follows directly from them:

- Rerun on the complete attack set and report false acceptance and false rejection rates, latency, confusion matrices, and ISO/IEC 30107-3 metrics with confidence intervals.
- Replace the histogram PAD with CNN-based PAD and add video or challenge–response liveness, so that replays and masks are addressed.
- Replace ORB with minutiae-based fingerprint matching and test with a validated multi-sample dataset.
- Use physical spoof samples, such as printed photographs, screen replays, and silicone or gelatin fingers.
- Adopt keyed hashing for credentials and add replay and transport protections.
- Tune each gate threshold against both unauthorized acceptance and false rejection to balance security and usability.

A. Towards a Deployable System

Moving from this prototype to something that could be trialled in the field would require work on four fronts. The first is *evidence*: a complete evaluation on a large, public benchmark with physical spoof samples, reported with confidence intervals and per-group error rates. The second is *sensing*: replacing static image upload with a capture step that records short video from a trusted device, so that liveness can be checked from motion and the capture channel can be authenticated. The third is *operations*: monitoring the audit log for repeated failures at one gate, which is a signature of an attacker probing the system, and rate-limiting an identity number after repeated failed attempts. The fourth is *governance*: legal review under the Digital Personal Data Protection Act, a clear retention policy for biometric templates and logs, and an accessible fallback for people whose biometrics cannot be captured reliably, such as manual workers with worn fingerprints, so that a strict security rule does not become a barrier to service.

IX. Conclusion

We built Identity Shield, a five-gate sequential authentication pipeline with a working web portal, which converts the threats named in a 2024 Aadhaar security survey into an executable and measurable defense. Requiring every gate to pass limits the chance that an attack is accepted to that of its weakest gate and removes the compensation weakness of score-level fusion. In a preliminary sampled run, the unauthorized acceptance rate fell from 100.0% for credential-only authentication to 9.0% for the full pipeline, and the 51.5% for a conventional identity-plus-matching design shows that most of the improvement comes from adding presentation-attack detection. Fingerprint attacks were the least well detected (84%).

A. Summary of Findings

Three findings stand out. First, each added layer reduced unauthorized acceptance in the sampled run, from 100.0% to 76.5% to 51.5% and finally to 9.0%, so the layers are complementary and not redundant. Second, the largest single reduction came from the presentation-attack gates and not from the extra matchers, which shows that checking whether a sample is real matters as much as checking whose it is. Third, the weakest category, fingerprint attacks at 84% detection, matches the weakest components, a global-statistics PAD and a keypoint matcher not designed for ridge patterns, which shows where effort should go next.

B. Lessons Learned

Building the system taught the team four practical lessons. Simple, explainable models are enough to demonstrate an architecture, but they set a ceiling on how far the numbers can go. Server-side enforcement of the gate order is more important than any individual model, because a client-side order can be bypassed. Synthetic identities make the work safe and repeatable, but they also hide questions that only real, consented data could answer, such as how genuine users from different groups fare. And logging every attempt with its per-gate result turned out to be what made the experiments analyzable at all, since the charts in this paper come directly from that log.

The main lessons are that gate order and hard-stop semantics are a cheap and effective structural defense, and that the strength of the whole system is limited by its weakest gate. The results are preliminary, and the next step is a full evaluation with the missing metrics and stronger fingerprint gates.

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